

Title: Design and Implementation of an AI-Based Student Support Assistant using Google Dialogflow

1. Aim

To design and implement an AI-based Student Support Assistant using Google Dialogflow that provides instant responses to student queries related to fee payment, scholarships, examination results, faculty contact, and placement services.

2. Problem Analysis and Agent Design

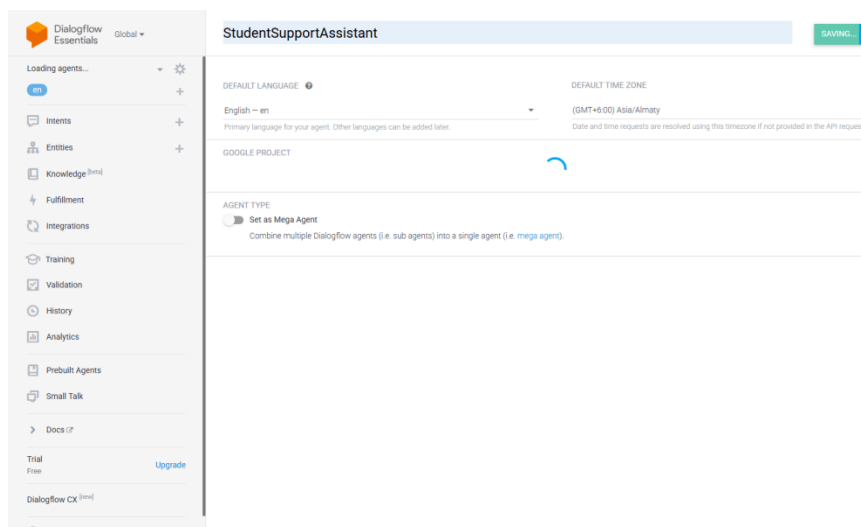
The university receives a large number of student queries every day regarding fee payment, scholarships, examination results, faculty contact, and placement services. Responding to these queries manually takes time and increases the workload of university staff. Students may also experience delays in getting accurate information, especially outside office hours. Therefore, an AI-based Student Support Assistant is required to provide instant, accurate, and 24/7 responses to student queries.

Why Artificial Intelligence?

Artificial Intelligence enables chatbots to understand natural language, identify user intentions, and provide relevant responses automatically. An AI-based chatbot can answer multiple student queries simultaneously, reduce the workload of university staff, improve response speed, and provide consistent information at any time. By using Dialogflow, the chatbot recognizes user intents and entities, making interactions more intelligent and user-friendly.

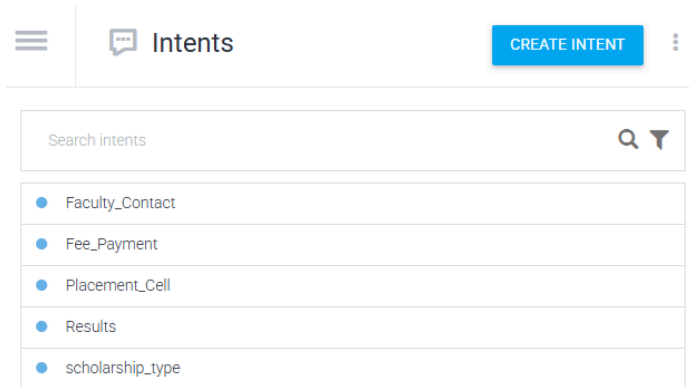
3. Dialogflow Agent Development

Step 1: Created Dialogflow Agent



Step 2:Created Intents

- Fee_Payment
- Scholarship_type
- Results
- Faculty_Contact
- Placement_Cell



Intents

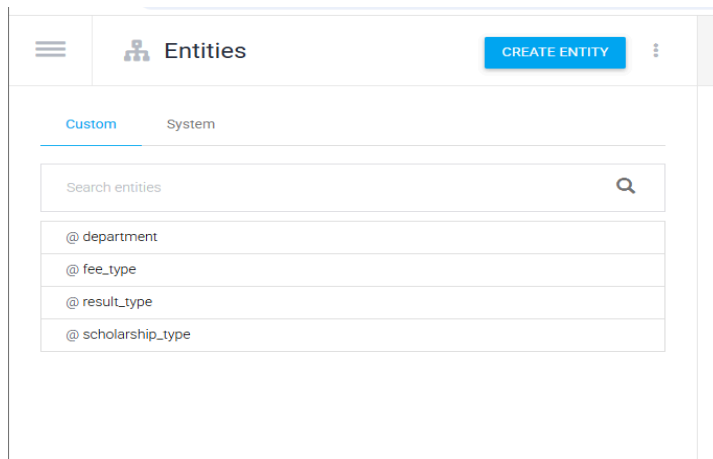
CREATE INTENT

Search intents

- Faculty_Contact
- Fee_Payment
- Placement_Cell
- Results
- scholarship_type

Step 3:Created Entities

Entity	Values
fee_type	Tuition, Hostel, Examination, Transport
scholarship_type	Merit, Sports, Minority
result_type	Semester, Mid, Final
department	CSE, AI, ECE, EEE



Entities

CREATE ENTITY

Custom System

Search entities

- @ department
- @ fee_type
- @ result_type
- @ scholarship_type

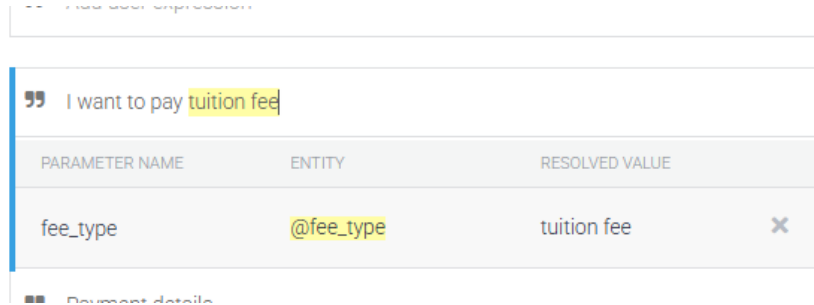
Step 4: Linked Entities to Intents

Example:

Training Phrase : I want to pay tuition fee

Highlighted: tuition fee

Entity: @fee_type



PARAMETER NAME	ENTITY	RESOLVED VALUE	
fee_type	@fee_type	tuition fee	X

Step 5: Responses

Add responses

Choose one intent (for example, **Fee_Payment**) and show its response.

Example Response (Fee_Payment):

You can pay your semester fees through the University Student Portal.

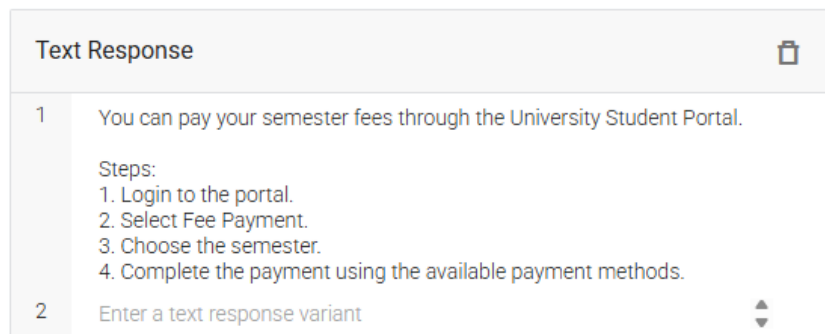
Steps:

1. Login to the Student Portal.
2. Select **Fee Payment**.
3. Choose the semester.
4. Complete the payment before the deadline.

Responses ?



DEFAULT +



	Text Response	
1	You can pay your semester fees through the University Student Portal. Steps: 1. Login to the portal. 2. Select Fee Payment. 3. Choose the semester. 4. Complete the payment using the available payment methods.	
2	Enter a text response variant	▲ ▼

Step 6: Contexts

Intent	Output Context
Fee_Payment	fee_payment_context
Scholarship	scholarship_context
Results	result_context
Faculty_Contact	faculty_context
Placement_Cell	placement_context

Example 1: Fee_Payment Intent

Output Context : fee_payment_context

Lifespan: 5

Contexts ?

Add input context

5 fee_payment_context

Add output context

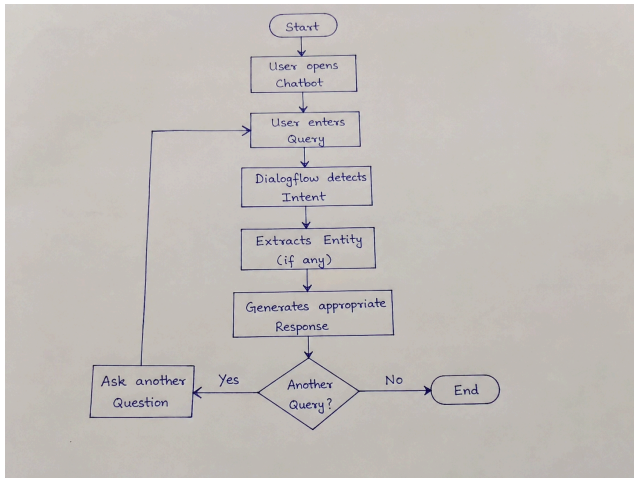
4. Problem Formulation and Conversation Flow

Initial State : User opens chatbot.

Goal State : Student receives the required information.

Actions

- Ask fee details
- Ask scholarship
- Ask result
- Ask faculty contact
- Ask placement details



5. Search Strategy Application

Breadth First Search (BFS)

- User enters query.
- Dialogflow checks all intents.
- Finds the closest matching intent.
- Generates the correct response.

Why BFS?

- Easy navigation.
- Fast intent selection.
- Suitable for small chatbot.

6. Testing and Evaluation

Create a table.

Test Case	User Input	Expected Output	Result
1	Hi	Welcome message	Pass
2	Fee payment	Fee response	Pass
3		Default fallback	Pass

Test Case	User Input	Expected Output	Result
	Weather today		

Try it now

Agent

USER SAYS

i want to pay tuition fee

COPY CURL

DEFAULT RESPONSE

You can pay your semester fees through the University Student Portal. Steps: 1. Login to the portal. 2. Select Fee Payment. 3. Choose the semester. 4. Complete the payment using the available payment methods.

CONTEXTS

fee_payment_context

RESET CONTEXTS

INTENT

Fee_Payment

Try it now

Agent

USER SAYS

What is the weather today?

COPY CURL

DEFAULT RESPONSE

Not available

CONTEXTS

scholarship_context

fee_payment_context

RESET CONTEXTS

INTENT

Not available

ACTION

Not available

7. Conclusion

The AI-based Student Support Assistant was successfully implemented using Google Dialogflow. The chatbot correctly identified user intents, extracted entities, and provided accurate responses to student queries. The chatbot can be further enhanced by integrating it with the university database for real-time information.